

Simplifying Ratios - No Decimals/Fractions

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Find the HCF for these numbers, which is the largest number that will divide into both numbers.

Examples

4, 6. = 2 9, 12. = 3

1 3, 6. = 3 2 2, 8. = 2

3 9, 12. = 3 4 5, 20. = 5

5 12, 14. = 2 6 9, 15. = 3

7 6, 12. = 6 8 15, 30. = 15

Now use the HCF to simplify these ratios. Use division.

9 15 : 10 $\frac{15 \div 5}{\text{HCF}} \frac{10 \div 5}{\text{HCF}}$ = 3 : 2

10 8 : 12 $\frac{8 \div 4}{\text{HCF}} \frac{12 \div 4}{\text{HCF}}$ = 2 : 3

11 14 : 35 $\frac{7}{\text{HCF}}$ = 2 : 5

12 24 : 18 $\frac{6}{\text{HCF}}$ = 4 : 3

13 10 : 100 $\frac{10}{\text{HCF}}$ = 1 : 10

Find the HCF then use it to simplify these ratios.

14 4 : 8 $\frac{\square}{\text{HCF}}$ = :

15 9 : 6 $\frac{\square}{\text{HCF}}$ = :

16 12 : 20 $\frac{\square}{\text{HCF}}$ = :

17 30 : 15 $\frac{\square}{\text{HCF}}$ = :

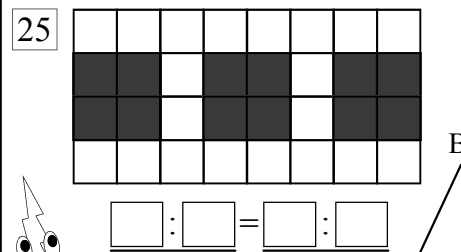
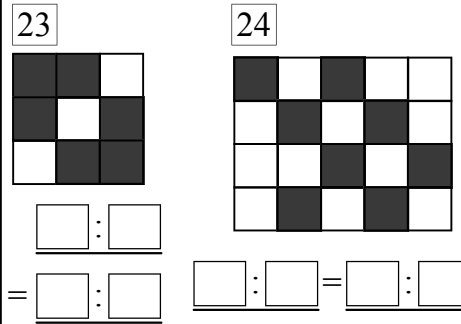
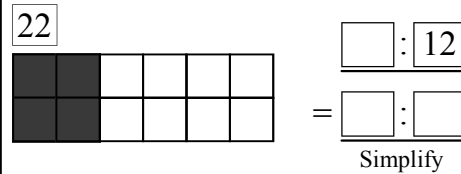
18 12 : 24 $\frac{\square}{\text{HCF}}$ = :

19 100 : 50 $\frac{\square}{\text{HCF}}$ = :

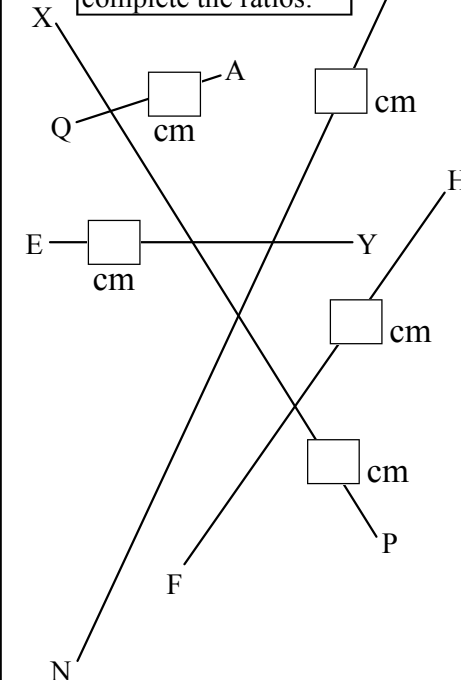
20 18 : 30 $\frac{\square}{\text{HCF}}$ = :

21 24 : 32 $\frac{\square}{\text{HCF}}$ = :

Find the non-simplified ratio for these grids, then simplify. Find the ratio for shaded : total.



Measure the lines then complete the ratios.



26 QA : PX = : = :

27 NB : HF = : = :

28 YE : XP = : = :

29 AQ : BN = : = :

30 FH : HF = : = :

Simplify these, you can use your calculator's fraction key. You have to solve triples mentally.

1st number is smaller than the 2nd number

Examples 8 : 12 = 2 : 3

Screen: '2/3' Calc: 8 [a^{b/c}] 12 [=]

A whole number answer is written ':1'

12 : 3 = 4 : 1

Screen: '4' Calc: 12 [a^{b/c}] 3 [=]

Change mixed numeral to improper fraction

Screen: '1 1/3' 12 : 9 = 4 : 3

Calc: 12 [a^{b/c}] 9 [=] $\frac{\text{SHIFT}}{\text{OR } 2^{\text{nd}} \text{ F}}$ [a^{b/c}]

31 16 : 48 = :

32 24 : 36 = :

33 30 : 6 = :

34 60 : 45 = :

35 18 : 20 = :

36 85 : 50 = :

37 36 : 27 = :

38 49 : 21 = :

39 100 : 95 = :

40 50 : 2 = :

41 500 : 50 = :

42 45 : 18 = :

43 4 : 12 : 8 = : :

44 9 : 6 : 3 = : :

45 25 : 5 : 20 = : :

46 16 : 8 : 24 = : :

47 10 : 50 : 100 = : :

48 20 : 40 : 50 = : :

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7 6, 12. = 6 8 15, 30. = 15

Now use the HCF to simplify these ratios. Use division.

9 15 : 10 $\frac{15 \div 5}{10 \div 5}$ = 3 : 2
HCF

10 8 : 12 $\frac{8 \div 4}{12 \div 4}$ = 2 : 3
HCF

11 14 : 35 $\frac{7}{7}$ = 2 : 5
HCF

12 24 : 18 $\frac{6}{6}$ = 4 : 3
HCF

13 10 : 100 $\frac{10}{10}$ = 1 : 10
HCF

Find the HCF then use it to simplify these ratios.

14 4 : 8 $\frac{4}{4}$ = 1 : 2
HCF

15 9 : 6 $\frac{3}{3}$ = 3 : 2
HCF

16 12 : 20 $\frac{4}{4}$ = 3 : 5
HCF

17 30 : 15 $\frac{15}{15}$ = 2 : 1
HCF

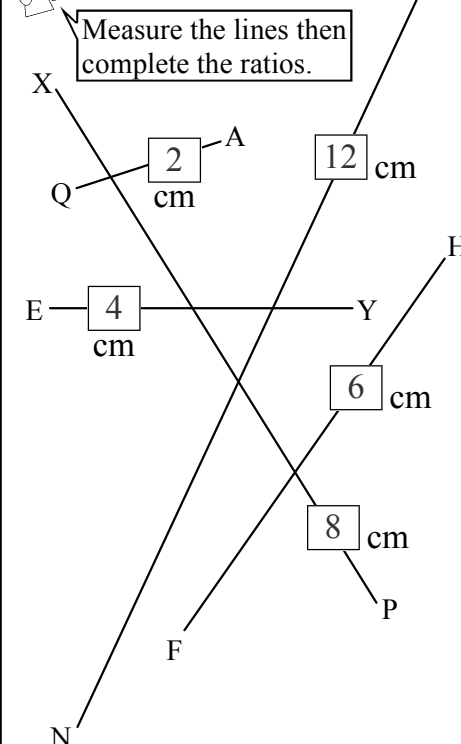
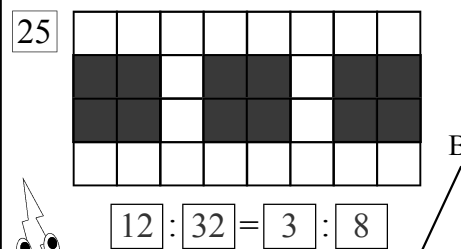
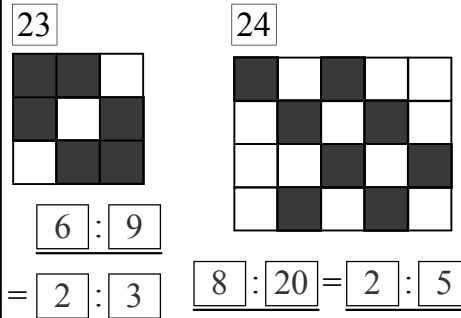
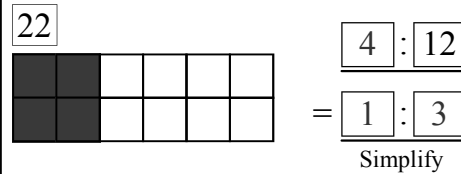
18 12 : 24 $\frac{12}{12}$ = 1 : 2
HCF

19 100 : 50 $\frac{50}{50}$ = 2 : 1
HCF

20 18 : 30 $\frac{6}{6}$ = 3 : 5
HCF

21 24 : 32 $\frac{8}{8}$ = 3 : 4
HCF

Find the non-simplified ratio for these grids, then simplify. Find the ratio for shaded : total.



26 QA : PX = $\frac{2}{8}$ = $\frac{1}{4}$

27 NB : HF = $\frac{12}{6}$ = $\frac{2}{1}$

28 YE : XP = $\frac{4}{8}$ = $\frac{1}{2}$

29 AQ : BN = $\frac{2}{12}$ = $\frac{1}{6}$

30 FH : HF = $\frac{6}{6}$ = $\frac{1}{1}$

Simplify these, you can use your calculator's fraction key. You have to solve triples mentally.

1st number is smaller than the 2nd number

Examples 8 : 12 = $\frac{2}{3}$: $\frac{3}{4}$
Screen: '2/3' Calc: 8 [a^{b/c}] 12 [=]

A whole number answer is written ':1'

12 : 3 = $\frac{4}{1}$: $\frac{1}{1}$
Screen: '4' Calc: 12 [a^{b/c}] 3 [=]

Change mixed numeral to improper fraction

Screen: '1 1/3' 12 : 9 = $\frac{4}{3}$: $\frac{3}{3}$
Calc: 12 [a^{b/c}] 9 [=] $\frac{4}{3}$ OR $\frac{4}{3}$ [a^{b/c}] 9 [=]

31 16 : 48 = $\frac{1}{3}$: $\frac{3}{9}$

32 24 : 36 = $\frac{2}{3}$: $\frac{3}{4.5}$

33 30 : 6 = $\frac{5}{1}$: $\frac{1}{1}$

34 60 : 45 = $\frac{4}{3}$: $\frac{3}{4.5}$

35 18 : 20 = $\frac{9}{10}$: $\frac{10}{10}$

36 85 : 50 = $\frac{17}{10}$: $\frac{10}{10}$

37 36 : 27 = $\frac{4}{3}$: $\frac{3}{4.5}$

38 49 : 21 = $\frac{7}{3}$: $\frac{3}{4.5}$

39 100 : 95 = $\frac{20}{19}$: $\frac{19}{19}$

40 50 : 2 = $\frac{25}{1}$: $\frac{1}{1}$

41 500 : 50 = $\frac{10}{1}$: $\frac{1}{1}$

42 45 : 18 = $\frac{5}{2}$: $\frac{2}{2.4}$

43 4 : 12 : 8 = $\frac{1}{3}$: $\frac{3}{4}$: $\frac{2}{3}$

44 9 : 6 : 3 = $\frac{3}{2}$: $\frac{2}{3}$: $\frac{1}{1}$

45 25 : 5 : 20 = $\frac{5}{1}$: $\frac{1}{1}$: $\frac{4}{5}$

46 16 : 8 : 24 = $\frac{2}{1}$: $\frac{1}{1}$: $\frac{3}{2}$

47 10 : 50 : 100 = $\frac{1}{5}$: $\frac{5}{10}$: $\frac{10}{100}$

48 20 : 40 : 50 = $\frac{2}{4}$: $\frac{4}{4}$: $\frac{5}{5}$